
Supplement to “No Randomized Speedup for Weakly Convex Optimization”

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A The Dead-Zone Chain

We use all definitions from Section 3 of the main paper. Let $q_0(t) = 0$ for $t \leq 0$, $q_0(t) = 3t^2 - 2t^3$ for $0 < t < 1$, and $q_0(t) = 1$ for $t \geq 1$. Thus $q_a(t) = q_0((t-a)/(1-a))$ with $a = 1/16$.

Lemma S1 (Smoothstep constants). *The function q_a is continuously differentiable, nondecreasing, and takes values in $[0, 1]$. Moreover,*

$$0 \leq q'_a \leq \frac{8}{5}, \quad \text{Lip}(q'_a) \leq \frac{512}{75}, \quad q_a(t) \leq \frac{256}{75} t_+^2.$$

For $\psi(u) = N^{-1} \sum_j q_a(Nu_j/4)$,

$$\|\nabla \psi(u)\| \leq \frac{2}{5} \sqrt{N}, \quad \text{Lip}(\nabla \psi) \leq \frac{32}{75} N.$$

Proof. On $(0, 1)$, $q'_0(t) = 6t(1-t)$ and $q''_0(t) = 6 - 12t$. Hence $\max q'_0 = 3/2$ and $\text{Lip}(q'_0) = 6$. Affine rescaling by $1-a = 15/16$ gives the first two constants. Since $q_0(s) \leq 3s_+^2$ and $((t-a)/(1-a))_+ \leq t_+/(1-a)$, the quadratic bound follows. The two bounds on ψ follow by the chain rule. $\square$

For $W = \{w \in \mathbb{R}_+^M : \|w\| \leq 1\}$ and $w_{M+1} = 0$, the support-function identity gives

$$\begin{aligned} \bar{f}(z) &= \max_{w \in W} F_w(z), \\ F_w(z) &= C(w) + \frac{2}{N} \sum_{m=1}^M w_m \phi(z^m) \\ &\quad - \frac{2}{N} \sum_{m=1}^M (1 - w_{m+1}) \psi(z^m). \end{aligned} \quad (1)$$

Lemma S2 (Analytic certificates). *The function $\bar{f}$ is one-Lipschitz, one-weakly convex, finite-valued, and bounded below. Its initial gap is at most $3M/N$.*

Proof. Each ϕ is convex and $N/8$ -Lipschitz. By Lemma S1, the negative progress part in (1) has weak-convexity modulus at most

$$\frac{2}{N} \frac{32N}{75} = \frac{64}{75} < 1.$$

The blocks are disjoint and the pointwise maximum preserves a common weak-convexity modulus. The tilted part has Lipschitz constant at most $\|w\|/4 \leq 1/4$. The progress part has gradient norm at most

$$\frac{2}{N} \sqrt{M \frac{4N}{25}} = \frac{4}{5} \sqrt{\frac{M}{N}} \leq \frac{4}{5\sqrt{12}}.$$

Their sum is below one.

The first term in the direct definition of $\bar{f}$ is nonnegative and every ψ lies in $[0, 1]$. Thus $\bar{f} \geq -2M/N$. At zero, only the first phase is positive and $\bar{f}(0) < 2/N$. Since $M \geq 2$, the initial gap is below $2(M+1)/N \leq 3M/N$. $\square$

Lemma S3 (Exact robust locality). *Let $\tau = 1/(64N^2)$. If $|z_{m,j}| < \tau$ for every coordinate with $\ell(m, j) > t$, then there is a neighborhood $\mathcal{N}$ of z such that*

$$\bar{f}(y) = \bar{f}(P_{t+1}y), \quad y \in \mathcal{N}.$$

Proof. Write

$$k_m = \min\{N, \max\{0, t - (m-1)L + 1\}\}.$$

Coordinates $j > k_m$ are more than level t and have magnitude below τ . Since $(N/4)\tau < 1/16 = a$, the function $q_a(Nu/4)$ is identically zero on a neighborhood of each such coordinate. Hence every progress term locally depends only on indices at most k_m .

Suppose $1 \leq k_m < N$. For $j \geq k_m + 2$, compare hidden affine facets:

$$\begin{aligned} &\left(c_{k_m+1} - \frac{N}{8} z_{k_m+1}^m\right) - \left(c_j - \frac{N}{8} z_j^m\right) \\ &\geq \frac{1}{16N} - \frac{N\tau}{4} = \frac{15}{256N} > 0. \end{aligned}$$

Thus only facet $k_m + 1$ can be active among hidden facets, and the strict margin persists locally. The fan $\phi(z^m)$ depends only on indices at most $k_m + 1$.

Now suppose $k_m = 0$ and $m \geq 2$. Then $k_{m-1} \leq L$. Hidden progress coordinates contribute zero, so

$\psi(z^{m-1}) \leq L/N \leq 3/16$. Also $\phi(z^m) \leq c_1 + N\tau/8$. Therefore

$$\sigma_m(z) \leq -\frac{1}{16N} + \frac{N\tau}{8} = -\frac{31}{512N} < 0.$$

The score remains negative locally, so the positive-part norm does not depend on block m . Its negative progress term is locally constant. If $m = 1$, then $t = -1$. The first facet is uniquely largest, by the same strict facet comparison, and it belongs to level zero.

Fully revealed blocks already lie in V_t . Combining the cases shows that every term in $\bar{f}$ is locally independent of coordinates beyond V_{t+1} , which proves the identity. $\square$

Lemma S4 (Phase and sign certificate). *Let $b = q_a(3/4) = 2783/3375$. If $\psi(z^M) < b$, then $R(z) = \|\sigma(z)_+\| > 0$. If $R(z) > 0$, every $g \in \partial \bar{f}(z)$ is coordinatewise nonpositive, $\|g\| \geq 1/(4\sqrt{N})$, and*

$$g_j^m < 0, z_j^m > 0 \implies z_j^m < \frac{13}{2N}.$$

Proof. Set $p_0 = 1$ and $p_m = \psi(z^m)$. Let $\hat{m}$ be the first index with $p_{\hat{m}} < b$. Then $p_{\hat{m}-1} \geq b$. Since q_a is nondecreasing, $p_{\hat{m}} < q_a(3/4)$ implies that $Nz_{\hat{m}}^m/4 < 3/4$ for some j . Since $c_j \geq 1$,

$$\phi(z^{\hat{m}}) > 1 - \frac{3}{8} = \frac{5}{8}.$$

It follows that $\sigma_{\hat{m}}(z) > 5/8 + b - 5/4 = b - 5/8 > 0$, proving $R(z) > 0$.

When $R(z) > 0$, the maximizing weight in (1) is unique:

$$w_m = \frac{[\sigma_m(z)]_+}{R(z)}, \quad \sum_m w_m^2 = 1.$$

Every subgradient has blocks

$$g^m = \frac{2}{N} (w_m u_m - (1 - w_{m+1}) \nabla \psi(z^m)),$$

$$u_m \in \partial \phi(z^m).$$

If $w_m > 0$, then $\sigma_m > 0$ and $\phi(z^m) > 5/4 - \psi(z^{m-1}) \geq 1/4$. The zero branch of ϕ is inactive. Consequently, $u_m \leq 0$ coordinatewise and $\|u_m\|_1 = N/8$, which gives $\|u_m\| \geq \sqrt{N}/8$. The progress gradient is nonnegative. Both terms in g^m are therefore nonpositive and cannot cancel. Hence

$$\|g^m\| \geq \frac{w_m}{4\sqrt{N}}, \quad \|g\|^2 \geq \frac{1}{16N} \sum_m w_m^2 = \frac{1}{16N}.$$

If $g_j^m < 0$, either an active affine facet contributes or $[\nabla \psi(z^m)]_j > 0$. In the first case, its value is above $1/4$, so

$$c_j - \frac{N}{8} z_j^m > \frac{1}{4} \implies z_j^m < \frac{8}{N} \left(\frac{17}{16} - \frac{1}{4} \right) = \frac{13}{2N}.$$

In the second case, $Nz_j^m/4 < 1$, so $z_j^m < 4/N$. This proves (S4). $\square$

Lemmas S2, S3, and S4 prove the robust-chain proposition in the main paper.

B Soft-Projection Calculus

Define $s_S(x) = \alpha(x)x$ with $\alpha(x) = (1 + \|x\|^2/S^2)^{-1/2}$.

Lemma S5 (Soft projection). *The map s_S has norm below S , is one-Lipschitz, and its Jacobian is $6/S$ -Lipschitz in operator norm. If h is one-Lipschitz and one-weakly convex, then $h \circ U^\top s_S$ is $(1 + 6/S)$ -weakly convex for every column-orthonormal U .*

Proof. Direct differentiation gives

$$J_s(x) = \alpha I - \frac{\alpha^3}{S^2} x x^\top.$$

Its eigenvalues are α in tangential directions and α^3 in the radial direction. They lie in $(0, 1]$, proving one-Lipschitzness. Differentiating J_s in a unit direction and applying the triangle inequality gives

$$\|DJ_s(x)\|_{\text{op}} \leq \frac{1}{S} \left(3\alpha^3 \frac{\|x\|}{S} + 3\alpha^5 \frac{\|x\|^3}{S^3} \right) \leq \frac{6}{S}.$$

Set $A(x) = U^\top s_S(x)$. For $g \in \partial h(A(x))$, weak convexity gives

$$h(A(y)) \geq h(A(x)) + \langle g, A(y) - A(x) \rangle - \frac{1}{2} \|A(y) - A(x)\|^2.$$

The first-order Taylor remainder of A has norm at most $3\|y - x\|^2/S$. Since $\|g\| \leq 1$ and A is one-Lipschitz, the right side is at least

$$h(A(x)) + \langle J_A(x)^\top g, y - x \rangle - \frac{1 + 6/S}{2} \|y - x\|^2.$$

This is the subgradient characterization of $(1 + 6/S)$ -weak convexity. $\square$

Lemma S6 (Wrapper class). *For $B = 512\sqrt{M}$ and $S = 16B$, the function*

$$H_U(x) = \frac{1}{4} \left[\bar{f}(U^\top s_S(x)) + 3(\sqrt{B^2 + \|x\|^2} - B) \right]$$

is one-Lipschitz, one-weakly convex, bounded below, and has initial gap at most $3M/(4N)$.

Proof. The first term before scaling is one-Lipschitz by composition. The radial term is convex, nonnegative, and three-Lipschitz. Hence the factor $1/4$ gives one-Lipschitzness. Lemma S5 gives weak-convexity modulus at most $(1 + 6/S)/4 < 1$. The radial term vanishes at zero, so adding it can only raise the infimum. Lemma S2 gives the gap bound. $\square$

C No Stationarity Created by the Wrapper

Lemma S7 (Wrapper stationarity). *Suppose $|\langle u_{M,j}, s_S(x) \rangle| < \tau$ for every $j > r_N := \lfloor N/2 \rfloor$. Then*

$$\|\nabla(H_U)_{1/2}(x)\| > \frac{1}{64\sqrt{N}}.$$

Proof. Suppose otherwise. Let $z = \text{prox}_{\frac{1}{2}H_U}(x) := \arg \min_y \{H_U(y) + \|y - x\|^2\}$ and $d = 2(x - z) = \nabla(H_U)_{1/2}(x)$. Proximal optimality gives $d \in \partial H_U(z)$. The derivative of $U^\top s_S$ is surjective, so the exact sub-differential chain rule yields some $g \in \partial \bar{f}(U^\top s_S(z))$ such that

$$4d = J_s(z)Ug + \frac{3z}{\sqrt{B^2 + \|z\|^2}}. \quad (2)$$

Because $\|J_s(z)Ug\| \leq 1$, if $\|z\| \geq B$ then

$$\|4d\| \geq \frac{3}{\sqrt{2}} - 1 > 1.$$

This contradicts $\|4d\| \leq 1/(16\sqrt{N})$. Thus $\|z\| < B$.

Set $\alpha = (1 + \|z\|^2/S^2)^{-1/2}$, $w = U^\top z$, and $y = \alpha w$. Since s_S is one-Lipschitz,

$$\begin{aligned} \|(y_{r_N+1}^M, \dots, y_N^M)\| &\leq \sqrt{N}\tau + \|s_S(z) - s_S(x)\| \\ &\leq \frac{1}{64N^{3/2}} + \frac{1}{128\sqrt{N}} \leq \frac{1}{64\sqrt{N}}. \end{aligned}$$

Using Lemma S1,

$$\begin{aligned} \psi(y^M) &\leq \frac{r_N}{N} + \frac{1}{N} \frac{256}{75} \frac{N^2}{16} \sum_{j>r_N} (y_j^M)^2 \\ &\leq \frac{1}{2} + \frac{1}{19200} < \frac{2783}{3375} = b. \end{aligned}$$

Lemma S4 now gives $g \leq 0$ coordinatewise and $\|g\| \geq 1/(4\sqrt{N})$.

Projecting (2) by $U^\top$ gives

$$U^\top(4d) = \alpha g + cw, \quad c = \frac{3}{\sqrt{B^2 + \|z\|^2}} - \frac{\alpha^3 \langle w, g \rangle}{S^2}.$$

Here $\alpha > 255/256$, $\|w\| < B$, and $\|g\| \leq 1$. Thus

$$c > \frac{2}{B} - \frac{1}{256B} > 0, \quad c < \frac{3 + 1/256}{B}.$$

Let $I = \text{supp}(g)$. By Lemma S4,

$$\|(w_I)_+\| \leq \frac{13\sqrt{MN}}{2\alpha N} = \frac{13\sqrt{M}}{2\alpha\sqrt{N}}.$$

Since $c > 0$, negative entries of w_I reinforce g_I . Therefore

$$\begin{aligned} \|4d\| &\geq \|\alpha g + cw\| \\ &\geq \frac{\alpha}{4\sqrt{N}} - \frac{3 + 1/256}{512\sqrt{M}} \frac{13\sqrt{M}}{2\alpha\sqrt{N}} \\ &> \frac{1}{5\sqrt{N}}. \end{aligned}$$

This gives $\|d\| > 1/(20\sqrt{N}) > 1/(64\sqrt{N})$, a contradiction. $\square$

Lemmas S6 and S7 prove the wrapper proposition in the main paper.

D The Conditional-Haar Reduction

We state the random-rotation argument in the exact local-oracle form used here. Let $\pi(1), \dots, \pi(n)$ order coordinates by nondecreasing level. Let $K_t = \{k : \ell(\pi(k)) = t\}$.

Lemma S8 (Adapted conditional-Haar bound). *Draw $U = (u_1, \dots, u_n)$ uniformly from the Stiefel manifold and let ξ be independent of U . Suppose v_k is measurable with respect to $\xi, u_1, \dots, u_{k-1}$ and $\|v_k\| \leq S$ for every $k \in [n]$. If*

$$d \geq n + 2S^2\tau^{-2} \log(2n^2/\delta),$$

then, with probability at least $1 - \delta$,

$$|\langle u_j, v_k \rangle| < \tau, \quad 1 \leq k \leq j \leq n.$$

Proof. Fix $k \leq j$ and condition on $\mathcal{G}_{k-1} := \sigma(\xi, u_1, \dots, u_{k-1})$. The remaining ordered frame is Haar in the orthogonal complement E_k of these columns, whose dimension is $d - k + 1$. In particular, u_j has a uniform marginal on the unit sphere in E_k . The vector v_k is fixed under the conditioning. If a_k is its projection onto E_k , then $\|a_k\| \leq S$. The standard spherical-cap bound, also used by Carmon et al. (2020, Appendix B.3), gives

$$\begin{aligned} \Pr(|\langle u_j, v_k \rangle| \geq \tau \mid \mathcal{G}_{k-1}) &\leq 2 \exp\left(-\frac{(d-k)\tau^2}{2S^2}\right) \\ &\leq 2 \exp\left(-\frac{(d-n)\tau^2}{2S^2}\right) \\ &\leq \frac{\delta}{n^2}. \end{aligned}$$

The event is impossible when $a_k = 0$. Averaging removes the conditioning. A union bound over fewer than n^2 pairs (k, j) proves the claim. Notice that no conditioning on preceding good events is needed because adaptedness holds globally. $\square$

Lemma S9 (Virtual-query rotation). *Fix a randomized algorithm with random seed ξ . Through the hard horizon, with probability at least $1 - \delta$, its effective query $y_t = s_S(x_t)$ satisfies*

$$|\langle u_{\pi(k)}, y_t \rangle| < \tau \quad \text{whenever } \ell(\pi(k)) \geq t,$$

provided the dimension bound in Lemma S8 holds.

Proof. We first define a surrogate transcript on every history. At round t , feed the algorithm the value and full subdifferential of the local wrapped function obtained by replacing $\bar{f}(y)$ with $\bar{f}(P_t y)$. This response depends only on x_t and columns through level t . Recursively, the surrogate query $\tilde{x}_t$ and its effective point $\tilde{y}_t = s_S(\tilde{x}_t)$ are measurable with respect to ξ and columns in levels below t .

Extend the algorithm after its output by fixed zero queries, so its surrogate queries are defined through the hard horizon. Construct a length- n virtual sequence. If $k \in K_t$ and $t < H$, set $v_k = \tilde{y}_t$. Set all later entries to zero. Denote the ordered column $u_{\pi(k)}$ by $\tilde{u}_k$. For $k \in K_t$, the point v_k depends only on columns in levels below t , hence only on $\xi, \tilde{u}_1, \dots, \tilde{u}_{k-1}$. Repetition within a layer does not use the same-layer columns already inserted in the virtual order. Lemma S8, applied to the reordered frame, now gives the claimed inequalities for the surrogate points.

It remains to couple the surrogate and true transcripts. At round zero their queries agree because both depend only on ξ . Suppose they agree through the query at round t . The adapted-Haar event makes every coordinate of $U^\top s_S(x_t)$ at level at least t smaller than τ . Applying Lemma S3 with index $t - 1$ shows that $\bar{f}$ agrees locally with $\bar{f} \circ P_t$. Values and full subdifferentials are local quantities. The exact subdifferential chain rule through $U^\top s_S$ therefore makes the true oracle response equal to the surrogate response. The soft projection and radial term are fixed and reveal no additional column. The next queries agree by induction. Thus the surrogate inequalities hold for the true queries through the hard horizon. $\square$

Lemma S9 proves the random-rotation lemma in the main paper. Notice that local equality is what controls the entire subdifferential set. A norm approximation would not suffice.

E Scaling and Completion

Lemma S10 (Exact scaling). *If H is one-Lipschitz and one-weakly convex, define*

$$F(x) = \frac{G^2}{\rho} H\left(\frac{\rho}{G}x\right).$$

Then F is G -Lipschitz and ρ -weakly convex, and

$$\nabla F_{1/(2\rho)}(x) = G \nabla H_{1/2}\left(\frac{\rho}{G}x\right).$$

Proof. The Lipschitz and weak-convexity claims follow directly by substitution. For $y = \rho x/G$, changing variables in the proximal problem gives

$$\text{prox}_{\frac{1}{2\rho}F}(x) = \frac{G}{\rho} \text{prox}_{\frac{1}{2}H}(y).$$

Multiplying the displacement by 2ρ proves the gradient identity. $\square$

Proof of Theorem 1. Set $\kappa = 1/64$ and

$$r = \min\{\rho\Delta/G^2, 1\}, \quad N = \left\lfloor \frac{\kappa^2 G^2}{\varepsilon^2} \right\rfloor, \quad M = \left\lfloor \frac{Nr}{12} \right\rfloor.$$

It is enough to assume $\varepsilon^2 \leq c_0 \min\{G^2, \rho\Delta\}$ with $c_0 = \kappa^2/120$. Then $N \geq 60$, $Nr \geq 60$, and $2 \leq M \leq N/12$. Let F_U be the scaled wrapper. Lemma S6 and Lemma S10 show that it is G -Lipschitz and ρ -weakly convex. Its initial gap is at most

$$\frac{G^2}{\rho} \frac{3M}{4N} \leq \frac{G^2 r}{16\rho} \leq \Delta.$$

Let $H = (M - 1)L + \lfloor N/2 \rfloor$. Since $N \geq 60$, $L \geq N/6$, and therefore $H \geq MN/120$. Apply Lemma S9 with $\delta = 1/4$. Given an algorithm for F_U , rescaling its queries by ρ/G and its oracle transcript by the inverse factors in Lemma S10 gives an induced algorithm for H_U . On its good event, every query before H satisfies the terminal projection condition of Lemma S7. Exact scaling gives

$$\|\nabla(F_U)_{1/(2\rho)}(x_t)\| > \frac{\kappa G}{\sqrt{N}} \geq \varepsilon.$$

Index the oracle calls as $x_0, \dots, x_{T-1}$. After $T < H$ calls, append the transcript-measurable output as x_T . It is still before level H , so the same statement applies to it.

The probability of the good event, averaged jointly over U and algorithmic randomness, is at least $3/4$. Hence some fixed frame U makes the algorithm fail with probability at least $3/4$, which rules out success probability $1/2$. Finally, $N \geq \kappa^2 G^2/(2\varepsilon^2)$ and $M \geq Nr/24$, so

$$H \geq \frac{MN}{120} \geq \frac{\kappa^4}{11520} \frac{G^2 \min\{\rho\Delta, G^2\}}{\varepsilon^4}.$$

Reducing the numerical constant accounts for the appended output query. This proves the theorem. $\square$

References

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